

1. Administrative & Study Identification

Full Study Title: A Study of Subcutaneous (SC) Pembrolizumab Coformulated With Berahyaluronidase Alfa (MK-3475A) vs Intravenous Pembrolizumab in Adult Participants With Metastatic Non-small Cell Lung Cancer (NSCLC) (MK-3475A-D77)-Japan Extension
Short Title/Acronym: MK-3475A-D77 Japan Extension **Protocol Number:** MK-3475A-D77 **NCT Number:** NCT06212752 **Posting ID:** 324789051674853911 **Sponsor/Company Name:** Merck **Investigational Product:** Pembrolizumab Coformulated With Berahyaluronidase Alfa (MK-3475A) SC / Pembrolizumab IV **Phase of Development:** Phase 3 (Criteria Type: PHASE_REQUIREMENT - Trial must be in PHASE3)
Trial Status: Active, not recruiting (ACTIVE_NOT_RECRUITING)
Medical Monitor: [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the pharmacokinetics (PK) and safety of subcutaneous (SC) MK-3475A versus intravenous (IV) pembrolizumab, administered with chemotherapy in the first-line treatment of adult Japanese participants with metastatic non-small cell lung cancer (NSCLC). The primary hypotheses of this study are that pembrolizumab (+) berahyaluronidase alfa subcutaneous (SC) is noninferior to pembrolizumab IV with respect to PK parameters. * **Secondary Objectives:** To evaluate the comparable clinical efficacy, overall survival (OS), and progression-free survival (PFS) of the SC formulation versus standard IV infusion.

2.2 Background & Rationale To address the need for a more efficient and convenient delivery method for immunotherapy in metastatic NSCLC patients. While IV pembrolizumab is highly effective, it requires longer infusion times. Formulating pembrolizumab with berahyaluronidase alfa allows for a subcutaneous injection with a median administration time of approximately two minutes, potentially reducing healthcare resource utilization and patient burden while maintaining the same pharmacokinetic exposure and clinical benefits. *Study Extension Details:* The Japan extension study will require approximately six years which includes one additional year (beyond the global study's last participant last study related contact) from the time the first participant provides informed consent until the last participant's last study related contact. As of Amendment 1 of the supplemental statistical analysis plan (effective date: 23 Aug 2024), patient-reported outcomes will no longer be secondary outcome measures of the study.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Intravenous Pembrolizumab) * **Blinding:** Open-label (None) * **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration * **Target N\$:** Approximately 39 adult Japanese participants * **Number of Sites:** Multicenter (Japan Extension) * **Estimated Study Start:** [Date] * **Estimated Primary Completion:** [Date]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult Japanese participants with treatment-naïve metastatic NSCLC. 2. Has histologically or cytologically confirmed diagnosis of squamous or non-squamous Non-small Cell Lung Cancer (NSCLC). 3. Must provide an archival tumor tissue sample or a newly obtained core, incisional, or excisional biopsy of a tumor lesion not previously irradiated. 4. Has a life expectancy of at least 3 months.

4.2 Exclusion Criteria 1. Has a diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements. 2. Has received prior systemic anticancer therapy for metastatic NSCLC. 3. Has received prior systemic anticancer therapy including investigational agents within 4 weeks before randomization. 4. Has received prior radiotherapy within 2 weeks of start of study intervention or has radiation-related toxicity requiring corticosteroids. 5. Has received radiation therapy to the lung (>30 Gray) within 6 months of start of study intervention. 6. Has received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 7. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy. 8. Has a known additional malignancy that is progressing or has required active treatment within the past 3 years. 9. Has an active autoimmune disease that has required systemic treatment in past 2 years. 10. Has an active infection requiring systemic therapy. 11. Has a history of human immunodeficiency virus (HIV) infection. 12. Has a history of Hepatitis B or C. 13. Has not adequately recovered from major surgery or has ongoing surgical complications. 14. Has a history of allogenic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * **Arm 1:** MK-3475A (Pembrolizumab + Berahyaluronidase alfa) 790 mg SC injection on Day 1 of each 6-week cycle + Platinum Doublet Chemotherapy. * **Arm 2:** Pembrolizumab 400 mg IV infusion on Day 1 of each 6-week cycle + Platinum Doublet Chemotherapy. * **Route of Administration:** Subcutaneous (SC) for Arm 1; Intravenous (IV) for Arm 2 and Chemotherapies. * **Duration of Treatment:** 18 cycles (up to approximately 108 weeks).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Platinum doublet chemotherapy agents including Cisplatin (Drug Preferred Name: cisplatin, Trade Name: Platinol, ATC Code: L01XA01), Pemetrexed, Carboplatin, Paclitaxel, Nab-paclitaxel; Filgrastim, Pegylated filgrastim. * **Prohibited:** Other investigational therapies or non-approved concurrent anticancer treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy Collection
Baseline	Day 0	Randomization, Baseline PK Labs, First Dose (SC or IV)
Treatment Cycles	Every 6 Weeks (3)	Study Drug Administration (Q6W), Chemotherapy, Safety Labs, PK Assessments
End of Study	Week 108	Final Efficacy Assessment, Survival Follow-up, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Objective Response Rate (ORR):** Defined as the percentage of participants who have a confirmed complete response (CR: disappearance of all target lesions) or partial response (PR: At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters) per Response Evaluation Criteria In Solid Tumors 1.1 (RECIST 1.1) as assessed by blinded independent central review (BICR). The percentage of participants with CR or PR will be reported. * Pharmacokinetics (PK) equivalent measurements and Incidence/Severity of Adverse Events (AEs) to establish safety profile. * **Measurement Method:** Central Laboratory Analysis of blood samples, standard AE clinical grading, and BICR tumor assessment.

7.2 Secondary & Exploratory Endpoints * **Cycle 1:** Area Under the Curve From Time 0 to 6 Weeks (AUC₀₋₆ Weeks) of Pembrolizumab After the First Dose * **Cycle 3:** Trough Serum Concentration (C_{trough}) of Pembrolizumab at Steady State * **Cycle 1:** Maximum Serum Concentration (C_{max}) of Pembrolizumab After the First Dose * Progression-Free Survival (PFS)

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard patient reporting, clinician evaluation during Q6W visits, and PK lab monitoring.
- **Serious Adverse Events (SAEs):** Must be reported within 24 hours of investigator awareness.
- **Data Monitoring Committee (DMC):** Internal review mechanisms governed by the sponsor to continually track the safety of the SC formulation versus IV.

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for efficacy; Safety Set for all treated patients.
- **Sample Size Justification:** Scaled appropriately for an extension study evaluating PK and safety non-inferiority in the specific Japanese demographic.
- **Handling of Missing Data:** Last Observation Carried Forward (LOCF) for continuous variables and Multiple Imputation where appropriate.

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Routine onsite and remote monitoring visits by the sponsor to assure data integrity.
- **Audit Trail:** Data entered into validated eCRF systems with comprehensive query resolution and database locking procedures.

11. Study Identifiers & Governance

Primary ID: MK-3475A-D77 **Posting ID:** 324789051674853911 **Registry Number:** NCT06212752 **Sponsor/Lead Organization:** Merck **Study Title:** A Study of Subcutaneous (SC) Pembrolizumab Coformulated With Berahyaluronidase Alfa (MK-3475A) vs Intravenous Pembrolizumab in Adult Participants With Metastatic Non-small Cell Lung Cancer (NSCLC) (MK-3475A-D77)-Japan Extension **Official Regulatory Title:** A Phase 3 Randomized, Open-label Clinical Study to Evaluate the Pharmacokinetics and Safety of Subcutaneous Pembrolizumab Coformulated With Hyaluronidase (MK-3475A) Versus Intravenous Pembrolizumab, Administered With Chemotherapy, in the First-line Treatment of Participants With Metastatic Non-small Cell Lung Cancer

12. Clinical Framework (NSCLC Specific)

Disease Area: Metastatic Non-small Cell Lung Cancer **Condition Name:** Metastatic Non Small Cell Lung Cancer (Preferred Name: Metastatic non-small cell lung cancer) **Preferred UMLS Name:** Metastatic Non-Small Cell Lung Carcinoma **Semantic Type:** Neoplastic Process **Definition:** Stage IV includes: (Any T, Any N, M1a); (Any T, Any N, M1b). M1a: Lung cancer with separate tumor nodule(s) in a contralateral lobe; tumor with pleural nodules or malignant pleural (or pericardial) effusion. M1b: Distant metastasis. (AJCC 7th ed.) **MeSH Code:** D016938 **SNOMED ID:** 457721000124104 **Source Level:** 5 **Medical Methodology:** Platinum doublet chemotherapy + Immunotherapy **Optimization Target:** PK equivalence and safety parity between SC and IV administration

13. Study Procedures & Methodology

Management Pathway: Standard oncological clinical pathway for metastatic NSCLC **Education Protocol (HCP):** Injection protocol training for SC Berahyaluronidase alfa coformulation **Education Protocol (Patient):** Cycle adherence and AE monitoring reporting **Quality Audit Mechanism:** Close PK sampling audit and trial protocol compliance tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 108 weeks of treatment **Onsite Visit Cadence:** Every 6 weeks (Q6W) corresponding to treatment cycle Day 1 **Remote Monitoring Frequency:** As required per patient safety protocol **Checklist Review Interval:** Pre-infusion/injection evaluation at every cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				